Swiggy Data Analytics Project

End-to-End Data Pipeline using Microsoft Fabric, SQL, and Power BI

1. Business Scenario

Swiggy is one of India's largest online food delivery platforms, connecting customers, restaurants, and delivery partners across multiple cities.

Every day, thousands of orders are placed through the platform. This generates large volumes of operational data related to:

- Customers
- Restaurants
- Orders
- Delivery partners
- Payments and delivery performance

Swiggy wants to use this data to **analyse business performance, improve delivery efficiency, and understand customer behaviour**.

To achieve this, the company wants to build a **modern data analytics solution using Microsoft Fabric and Microsoft Power BI**.

You are hired as a **Data Analyst** to design an **end-to-end analytics solution** that transforms raw data into meaningful business insights.

2. Project Objective

The objective of this project is to build a **complete data analytics pipeline** that:

1. Stores raw Swiggy datasets in a **Lakehouse**
2. Cleans and validates the data using **SQL**
3. Models the data inside a **Data Warehouse**
4. Creates a **Power BI Semantic Model**
5. Builds interactive **Power BI dashboards**
6. Publishes the final report for business stakeholders

3. Technology Stack

The project will use the following tools:

Layer	Technology
Data Storage	Lakehouse
Data Modeling	Data Warehouse
Data Cleaning	SQL
Data Processing	Fabric Data Pipelines
Semantic Layer	Power BI Semantic Model
Visualization	Power BI Reports

4. Data Architecture

Students must create the following architecture in the Fabric workspace.

Step 1 – Create Workspace

Create a workspace named:

Swiggy Analytics Project

This workspace will contain:

- Lakehouse
- Data Warehouse
- Data Pipelines
- Power BI Semantic Model
- Power BI Report

Step 2 – Data Ingestion (Lakehouse)

Upload raw datasets to the **Lakehouse** inside a folder structure.

Example structure:

Files
  Raw_Data
    customers.csv
    restaurants.csv
    orders.csv
    delivery_partners.csv
    deliveries.csv

The Lakehouse will act as the **raw data storage layer**.

Step 3 – Data Cleaning and Validation (SQL)

Use SQL queries to clean and validate the raw data.

Typical tasks include:

- Removing duplicate records
- Handling missing values
- Correcting invalid date formats
- Validating order amounts
- Standardizing city names
- Checking referential integrity between tables

Examples of validation checks:

- Orders without customers
- Orders without restaurants
- Negative order amount
- Delivery time earlier than pickup time

Step 4 – Data Warehouse Modeling

Create a **Star Schema** inside the Data Warehouse.

Dimension Tables

- dim_date

- dim_restaurants
- dim_dish
- dim_location

Fact Table

- fact_orders

Step 5 – Semantic Model

Create a **Power BI Semantic Model** on top of the Data Warehouse.

Define:

- Table relationships
- Measures
- KPIs

Example measures:

- Total Orders
- Total Revenue
- Average Order Value
- Average Delivery Time
- Orders per Restaurant

Step 6 – Power BI Report

Create an interactive **Power BI dashboard** that provides insights for business users.

The report should be published inside the Fabric workspace.